

Module2Lab3

September 13, 2026

1 Problem 4.4: Temperature Categories

This notebook uses a loop together with an `if/elif/else` chain to assign a category to each temperature reading.

Temperature cutoffs: - **Cold:** below 15°C - **Warm:** 15°C through 25°C - **Hot:** above 25°C

```
[1]: temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```
for temperature in temperatures:
    if temperature < 15:
        category = "cold"
    elif temperature <= 25:
        category = "warm"
    else:
        category = "hot"

    print(f"{temperature}°C is {category}.")
```

10°C is cold.

16°C is warm.

23°C is warm.

28°C is hot.

14°C is cold.

22°C is warm.

30°C is hot.

1.1 Conclusion

The loop successfully classified every temperature in the list. Temperatures below 15°C were labeled cold, those from 15°C to 25°C were labeled warm, and anything above 25°C was labeled hot. This shows how a loop paired with a conditional statements can efficiently categorize data.